

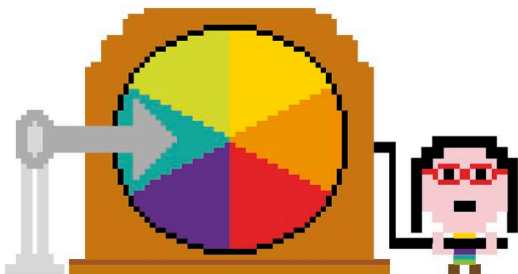
How it works

In this project, you'll use the `turtle` module and a clever looping technique to layer circles on top of each other in a spiral pattern. Every time a circle is drawn, the program slightly increases the parameters of the circle-drawing code. Each new circle is different from the last one drawn, making the pattern more interesting.

EXPERT TIPS

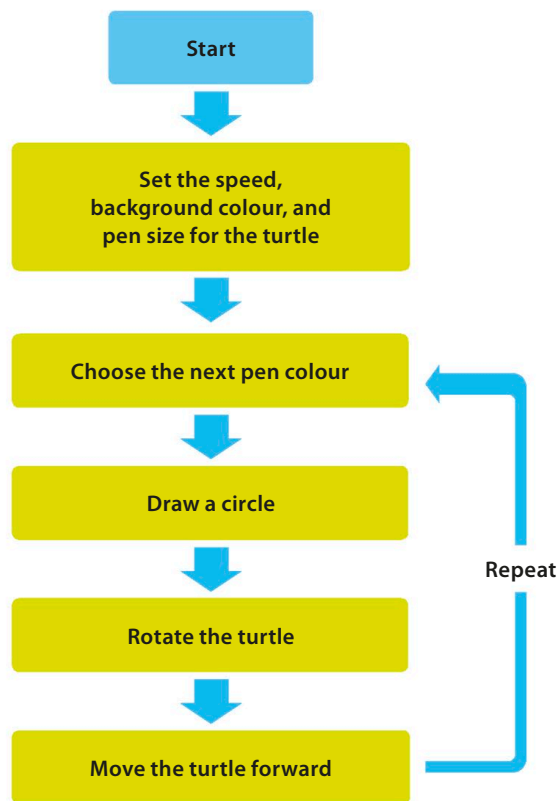
Cycling

To make the patterns colourful, this project uses a function called `cycle()` from the `itertools` module. The `cycle()` function allows you to cycle through a list of different colours over and over again. This makes it easy to use a different pen colour for each circle.



▽ Kaleido-spiral flowchart

The program sets some values that stay the same throughout, such as the turtle's speed, and then starts looping. The loop chooses a new pen colour, draws a circle, turns and moves the turtle, and then repeats itself. It stops when you quit the program.



Get drawing!

The first thing you'll draw on the screen is a simple circle. Next you'll repeat this circle, but with a slight change. Finally, you'll tweak the code to make the pattern more colourful and interesting.

1 Create a new file

Open IDLE and create a new file. Save it as "kaleido-spiral.py".

2 Import turtle

First you need to import the `turtle` module. This will be the main module you use. Type this line at the top of the program.

```
import turtle
```

Loads the entire `turtle` module